

Algebra II with Trigonometry

Chapter 6.3

Mental Math (GPT-5.4)

Saved tutoring response with MathJax rendering. Original PDF file:
[4531393_alg-2trig-h-chp-6-3-note.pdf](#)

1. **Basic Period** What is the period of $y = \sin x$?
2. **Cosine Pattern** Over one complete cycle, does $y = \cos x$ go through the pattern high-middle-low-middle-high or middle-high-middle-low-middle?
3. **Sine Pattern** Over one complete cycle, does $y = \sin x$ start at a maximum, minimum, or midline?
4. **Amplitude Check** What is the amplitude of $y = -4 \cos x$?
5. **Horizontal Stretch/Shrink** Compared with $y = \sin x$, is the graph of $y = \sin(3x)$ horizontally stretched or horizontally shrunk?
6. **Period from k** What is the period of $y = \cos(2x)$?
7. **Vertical Shift** For $y = 5 + 2 \sin x$, what is the vertical shift?
8. **Horizontal Shift** For $y = \sin(x - \pi/4)$, is the graph shifted left or right, and by how much?
9. **Range from Transformations** What is the range of $y = 3 + \sin x$?
10. **Vibration Model** For $v(t) = 0.7 \sin(880\pi t)$, what is the amplitude of the vibration?